# Andrei Silviu Dragnea

Professional automator. Strong believer in the power static analysis, refactoring and automating every aspect of the software development process. A developer eager to make other developers' lives easier.

## Education

**2013-2017** **BSc, Computer Science and Engineering**; Faculty of Automatic Control and Computers, Politehnica University of Bucharest (Bucharest)

*Thesis title: Automatic Recursion Removal in Java (as an* ***Intellij Plugin****)*

## Experience

- Intellij plugin **developer**
- Spring framework
- Async programming expert (Project Reactor from Spring, Kotlin coroutines, cats-effect from Scala, Rust async programming)
- Open source contributor to **intellij-kotlin**, **mockito**, **intellij-community**, **reactor-core**, **reactor-netty**, **netty**, **spring-framework**, Scala **kubernetes-client**.
- Proficient in Java, Kotlin, Scala, C and Python
- Author of an automatic Git **rebaser** written in Rust.
- Passionate about static analysis and automatic refactoring in Java, Kotlin, Scala and Rust
- Working on a **converter** from Java reactive code to Kotlin coroutines
- Experience with enterprise identity protocols and frameworks, such as OAuth2, SCIM2, SAML
- Worked with Kubernetes, Envoy, Grafana, Prometheus, AWS EC2, S3, Route53, Cloudwatch, Splunk, Datadog, NewRelic, Azure Active Directory
- Good knowledge of Operating Systems internals and how async frameworks are implemented
- Promoter of the **clean tests** paradigm

**March 2018 - May 2021** **Software Engineer at Adobe (Identity Management Services team)** I am part of the Identity Management Services team. I worked on implementing the server side of the SCIM2 protocol over a Spring 5 reactive stack. I developed a thin wrapper over Mockito and Spring TestContext framework in order to promote writing **clean tests**. I introduced Kotlin and coroutines to our team, as a developer-friendly alternative to writing non-blocking code. I made major software architectural changes by using smart refactoring tricks, resulting in a simpler system design.

**May 2021 - Present** **Software Engineer at Adobe (Media Analytics team)** I work on a performance-sensitive system collecting Analytics events from media players in the browser. I optimized the runtime performance by correctly handling blocking calls, thus avoiding starving the **cats-effect** compute pool. I also enforced strong static analysis checks on the Scala project at compile time, in order to maintain the high quality of the code. I am currently experimenting with re-writing the system in Rust, for improving the performance even more, while reducing the costs of running a JVM application in the cloud.

**September 2017 - December 2017** **Software Engineer at Instacar** I implemented a TCP server over a proprietary binary protocol for remote car control. I recompiled the Linux kernel with custom TCP settings in order to compensate for a faulty TCP implementation on the devices used for remote car control. I introduced type hints to the Python 3 codebase and I developed a Pycharm **plugin** for dead code elimination.

I always strive the find the simplest solution possible to a problem. I proactively fix problems from any software project that I encounter, whether it is internal to my company, or an open source project.


https://github.com/andreisilviudragnea
**andreisilviudragnea@gmail.com**